

V-MARCH: Multi-Agent Self-Verification with Visual Claim Regulation for VLM Hallucination

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Abstract

Hallucination remains a critical challenge for vision-language models (VLMs), especially in real-world scenarios where models generate image-inconsistent content. Existing mitigation methods typically rely on human annotations or external supervision, limiting their scalability to large-scale unlabeled image corpora. We propose **V-MARCH**, a reinforced visual self-check framework for reducing hallucination in VLMs without ground-truth labels. V-MARCH performs claim-level factual verification by decomposing model responses into atomic claims, transforming them into visual questions, and aggregating image-grounded verification signals as reinforcement learning rewards. To improve reward reliability, we further introduce a Visual Claim Regulation (VCR) mechanism that encourages informative, image-grounded descriptions while suppressing trivial or redundant claims. We evaluate V-MARCH on 12 benchmarks spanning hallucination evaluation (+3.30), real-world VQA (+1.74), and general multimodal understanding benchmarks (+4.07), where it consistently outperforms existing methods without human-annotated factual supervision. Moreover, V-MARCH generalizes effectively across model scales, demonstrating its potential as a scalable hallucination mitigation solution on large-scale real-world unlabeled image corpora.

1 Introduction

Vision-Language Models (VLMs) have demonstrated remarkable capabilities across a wide range of multimodal tasks, from image captioning to visual question answering (Achiam et al., 2023; Touvron et al., 2023; Liu et al., 2023b; Chen et al., 2024; Bai et al., 2025a; Comanici et al., 2025). However, hallucination remains a fundamental bottleneck: models frequently generate descriptions that are fluent and coherent yet factually inconsistent with the visual content of the image (Liu et al.,

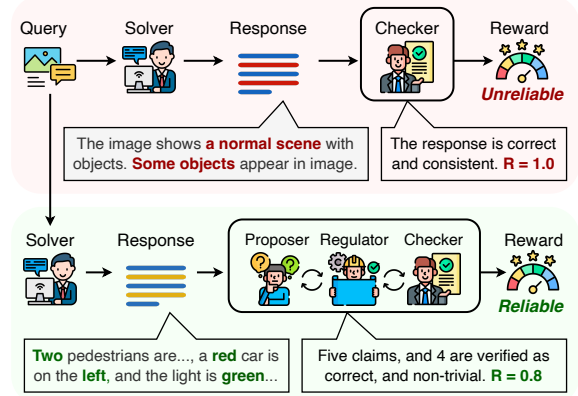


Figure 1: V-MARCH achieves reliable visual self-verification rewards through a multi-agent self-check mechanism, encouraging VLMs to generate more informative, image-supported, and diverse responses.

2024a; Rawte et al., 2025; Li et al., 2025a; Yang et al., 2025a). Such visual hallucinations manifest as fabricated objects, incorrect attributes, or erroneous spatial relationships, and represent a critical barrier to the reliable deployment of VLMs in real-world applications (Dahl et al., 2024; Kim et al., 2025). Reinforcement Learning from Verifiable Rewards (RLVR) has emerged as a promising paradigm for improving model factuality by optimizing directly against reward signals derived from output correctness (Shao et al., 2024; Yu et al., 2026; Wen et al., 2025; Yue et al., 2026). However, unlike domains such as mathematics or code generation where correctness can be programmatically verified, factual correctness in VLM generation is inherently difficult to verify from images. Existing approaches to mitigating VLM hallucination largely fall into two categories. The first relies on human annotations or external supervision, such as curated factual datasets, preference annotations, or learned reward models (Ouyang et al., 2022; Bai et al., 2022). While effective, these methods are expensive to scale and difficult to generalize to large-scale unlabeled image cor-

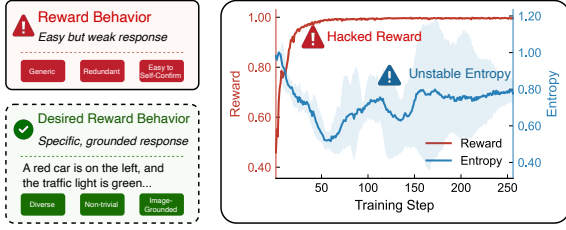


Figure 2: Motivation. Naive self-verification rewards vague but safe descriptions, causing rapid reward growth that reflects reward hacking rather than genuine visual grounding. Reliable visual self-verification should instead favor diverse, non-trivial, and image-grounded claims while maintaining stable training.

pora. The second category leverages self-checking or self-reflection mechanisms to improve factual consistency through iterative verification or critique (Madaan et al., 2023; Shinn et al., 2023; Li et al., 2026). However, they lack fine-grained image-grounded verification signals that can reliably assess whether generation is visually supported by the input image, as the verifier can easily rely on textual priors rather than visual evidence, leading to reward hacking and unstable training dynamics (Figure 2). A key open question remains: *How can VLMs obtain reliable and scalable image-grounded verification signals without relying on annotations or external supervision?*

To address this, we propose **V-MARCH**, a visual multi-agent self-check framework for scalable hallucination mitigation in VLMs. Our insight is that factual verification for VLMs can be approximated through visual regulated self-verification, where generated claims are validated solely against image-grounded visual evidence without accessing the original response (Madaan et al., 2023; Li et al., 2026). This multi-agent design reduces confirmation bias by decoupling generation from verification, forcing the verifier to rely on the image rather than inherited textual priors (Stechly et al., 2025; Jhaveri et al., 2026). Specifically, V-MARCH constructs an automated claim-level multi-agent factual verification pipeline for reinforcement learning. To further improve the reliability of self-verification rewards, we introduce a Visual Claim Regulation (VCR) mechanism, implemented as a **Regulator** agent that regularizes the rewards by encouraging informative and image-grounded descriptions while suppressing redundant or weakly supported claims. Experimental results across 12 benchmarks show that V-MARCH consistently outperforms existing methods, achieving sub-

stantial improvements without relying on ground-truth. Our main contributions are as follows:

- We propose **V-MARCH** method, the first visual multi-agent reinforced self-check framework for VLM hallucination mitigation.
- We introduce **VCR**, a novel mechanism that encourages diverse, informative, and image-grounded descriptions, thereby stabilizing self-verification rewards.
- Extensive experiments on 12 benchmarks demonstrate that V-MARCH consistently outperforms existing competitive baselines, establishing a scalable pathway for self-supervised VLM hallucination mitigation.
- Comprehensive ablating studies and analyses provide insights into the design of effective self-verification mechanisms for VLMs.

2 Related Work

VLM Hallucination. Hallucination in VLMs, where generated text is factually inconsistent with the visual input, has been studied along several mitigation strategies (Li et al., 2023; Wang et al., 2023). Supervised alignment methods adapt RLHF or DPO with human-annotated preference data to reduce hallucinated outputs (Sun et al., 2024; Yu et al., 2024; Fu et al., 2025; Yang et al., 2025c; Wu et al., 2025), but their reliance on costly labels limits scalability to large unlabeled corpora. Training-free inference-time approaches, such as contrastive decoding (Leng et al., 2024), attention-based penalty strategies (Huang et al., 2024), and dynamic correction decoding (Wang et al., 2025; Yin et al., 2025; Tang et al., 2025; Anand et al., 2026), intervene at generation without modifying model weights, yet leave underlying hallucination tendencies unchanged. Representation-editing methods (Jiang et al., 2025; Wang et al., 2026b) steer internal activations toward factual semantics with low overhead, but require explicit factual supervision to define the editing direction. Unlike these methods, V-MARCH constructs fine-grained rewards entirely from the model’s own outputs via a regulated self-verification pipeline, requiring no human annotation, no external verifier, and no corrupted-image heuristics, while the VCR mechanism further ensures reward reliability against redundant and image-agnostic claims.

3 Method

3.1 Problem Formulation

We consider reinforcement learning for vision-language generation without ground-truth labels. Let $\mathcal{D} = \{(\mathbf{I}, \mathbf{q})\}$ denote a collection of image-question pairs, where $\mathbf{I} = \{I_1, I_2, \dots, I_K\}$ is a set of input images and $\mathbf{q}$ is a textual instruction. A vision-language policy π_θ takes $x = (\mathbf{I}, \mathbf{q})$ as input and generates a response $\mathbf{y} \sim \pi_\theta(\cdot | \mathbf{I}, \mathbf{q})$. Our goal is to improve the factual grounding of π_θ by encouraging $\mathbf{y}$ to be consistent with the visual content of $\mathbf{I}$, thereby reducing hallucinated objects, attributes, and relations.

In the standard RLVR setting, a scalar reward $r(\mathbf{x}, \mathbf{y})$ can be computed by verifying the model output against a ground-truth answer or an executable checker. The policy is then optimized to maximize the expected reward while regularizing its deviation from a reference policy π_{ref} :

$$\max_{\pi_\theta} \mathbb{E}_{\mathbf{y} \sim \pi_\theta(\cdot | \mathbf{I}, \mathbf{q})} [r(\mathbf{I}, \mathbf{q}, \mathbf{y})] - \beta \text{KL}[\pi_\theta \| \pi_{\text{ref}}],$$

where β controls the strength of KL regularization, omitted in the following for simplicity.

We instantiate this objective with Group Relative Policy Optimization (GRPO) (Shao et al., 2024). For each input $x = (\mathbf{I}, \mathbf{q})$, GRPO samples a group of G responses from the rollout policy $\pi_{\bar{\theta}}$:

$$\{\mathbf{y}_i\}_{i=1}^G \sim \pi_{\bar{\theta}}(\cdot | \mathbf{I}, \mathbf{q}),$$

where each response receives a scalar reward $r_i = r(\mathbf{I}, \mathbf{q}, \mathbf{y}_i)$. GRPO then normalizes the rewards within the group to obtain the relative advantage $\hat{A}_i = (r_i - \mu_r) / (\sigma_r + \epsilon_{\text{std}})$, where μ_r and σ_r are the mean and standard deviation of the rewards in the group, and ϵ_{std} is a small constant for numerical stability. The policy is updated using the clipped surrogate objective $\mathcal{L}_{\text{GRPO}}(\theta)$:

$$-\mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \min(\rho_i \hat{A}_i, \text{clip}(\rho_i, 1 - \varepsilon, 1 + \varepsilon) \hat{A}_i) \right] \quad (1)$$

where $\rho_i = \frac{\pi_\theta(\mathbf{y}_i | \mathbf{I}, \mathbf{q})}{\pi_{\bar{\theta}}(\mathbf{y}_i | \mathbf{I}, \mathbf{q})}$, and $\varepsilon > 0$ is the clipping parameter.

The key challenge in our setting is that the reward $r(\mathbf{I}, \mathbf{q}, \mathbf{y})$ is not directly available. Unlike mathematical reasoning or code generation, where correctness can be verified by an exact answer or program execution, we cannot directly determine whether $\mathbf{y}$ faithfully describes the visual content of

$\mathbf{I}$. Existing self-verification approaches can be understood as constructing a surrogate reward R_{SV} to approximate the latent reward function $r(\mathbf{I}, \mathbf{q}, \mathbf{y})$. Following this perspective, we propose V-MARCH, a visual multi-agent self-verification framework that provides scalable fine-grained reward signals for annotation-free policy optimization.

3.2 V-MARCH Overview

V-MARCH adapts the Proposer–Checker paradigm (Wang et al., 2026a; Li et al., 2026) to the visual domain and extends it with a dedicated **Regulator** agent. The Regulator governs the quantity and quality of the claim set, addressing unreliable rewards to visual self-verification. As illustrated in Figure 3, V-MARCH orchestrates four agents in a sequential pipeline:

- **Solver.** Generates a response $\mathbf{y}$ conditioned on the image–instruction pair $(\mathbf{I}, \mathbf{q})$. Only the Solver is updated by GRPO training.
- **Proposer.** Decomposes $\mathbf{y}$ into a set of atomic factual claims $\mathcal{C} = \{c_1, \dots, c_m\}$ and converts each claim into a visual question-answer pair, yielding the raw claim set $\mathcal{Q} = \{(v_i, a_i)\}_{i=1}^m$.
- **Regulator.** Audits $\mathcal{Q}$ by (i) removing semantically redundant claims to prevent reward inflation via repetition, yielding the filtered set $\mathcal{Q}^* = \{(v_i, a_i)\}_{i=1}^n, n \leq m$; (ii) rewarding sufficient claim coverage via a claim-count signal r^{cnt} ; and (iii) penalizing image-agnostic claims that can be resolved without genuine visual perception via a quality signal r^{qual} . We refer to the claim-level quantity and quality control as **Visual Claim Regulation (VCR)**, which is described in detail in Section 3.4.
- **Checker.** Receives the original image $\mathbf{I}$ and questions $\{v_i\}$ from $\mathcal{Q}^*$, with no access to $\mathbf{y}$. It generates image-grounded reference answers $\{a'_i\}$ and verifies consistency with the claim-implied answers $\{a_i\}$ via majority voting, producing a consistency reward r^{cons} .

The final self-verifying reward aggregates signals from the Regulator and the Checker:

$$r^{\text{SV}}(\mathbf{I}, \mathbf{q}, \mathbf{y}) = \underbrace{r^{\text{cons}}}_{\text{Checker}} + \underbrace{r^{\text{cnt}} + r^{\text{qual}}}_{\text{Regulator}}. \quad (2)$$

This reward is used directly as the GRPO training signal for the Solver, requiring no human annotation or external supervision.

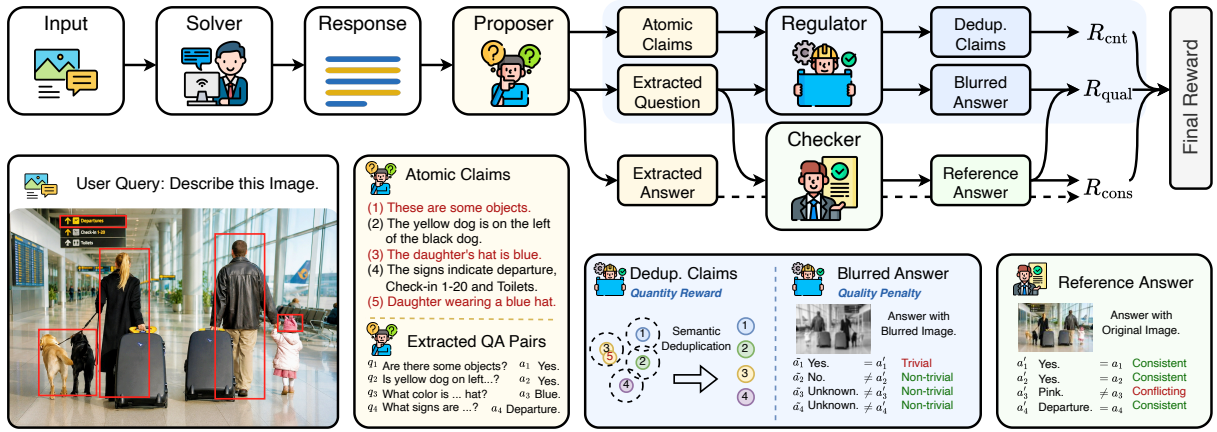


Figure 3: Overview of the V-MARCH pipeline. Given a **Solver**-generated response, the **Proposer** decomposes it into atomic factual claims and converts each into a visual question-answer pair. The **Regulator** performs Visual Claim Regulation (VCR) by deduplicating semantically redundant claims, rewarding claim quantity, and penalizing image-agnostic claims that can be answered without genuine visual perception. The **Checker** re-answers each question using the original image, and verifies consistency with the Solver’s answers. The final self-verifying reward aggregates the Checker’s consistency signal and the Regulator’s quantity and quality signals.

3.3 Proposer: Claim Extraction

Given y , the Proposer extracts a set of atomic factual claims:

$$\mathcal{C} = \{c_1, \dots, c_m\} = \text{Extract}(y), \quad (3)$$

where each c_i is a short, self-contained statement about a single visual fact (e.g., object existence, color, quantity, or spatial relation). Each claim is then converted into a visual question-answer pair:

$$(v_i, a_i) = \text{QA}(c_i), \quad i = 1, \dots, m, \quad (4)$$

where a_i is the answer implied by the claim. The resulting raw claim set $\mathcal{Q} = \{(v_i, a_i)\}_{i=1}^m$ is passed to the Regulator for quality control.

Example. For the query “Describe this image” applied to an airport scene, the Solver might generate a response y . The Proposer then extracts atomic claims and converts each to a QA pair, for instance: “The yellow dog is on the left of the black dog” $\rightarrow$ (Q: Is the yellow dog on the left of the black dog?, A: Yes); “The daughter’s hat is blue” $\rightarrow$ (Q: What color is the daughter’s hat?, A: Blue).

3.4 Regulator: Visual Claim Regulation

The Regulator addresses degenerate behaviors that undermine the reliability of visual self-verification if left unchecked: (1) a model can repeat near-identical claims to inflate r^{cons} ; (2) it can produce easily verified claims to obtain a stable reward while remaining under-informative; (3) it can generate image-agnostic claims (e.g., “the image con-

tains objects”) whose answers follow from language priors rather than genuine visual perception. The Regulator counteracts these through semantic deduplication and complementary reward signals.

Semantic Deduplication. Removing semantically overlapping claims from $\mathcal{Q}$ using cosine similarity over sentence embeddings $\phi(\cdot)$:

$$\text{Sim}(c_i, c_j) = \frac{\phi(c_i)^\top \phi(c_j)}{\|\phi(c_i)\| \cdot \|\phi(c_j)\|}. \quad (5)$$

Pairs exceeding a type-aware similarity threshold are merged, yielding the deduplicated claim set $\mathcal{Q}^* = \{(v_i, a_i)\}_{i=1}^n, n \leq m$.

Example. Continuing the airport-scene example, the claim “Daughter wearing a blue hat” is identified as a near-duplicate of “The daughter’s hat is blue” and is removed, reducing the claim set from $m = 5$ to $n = 4$ deduplicated claims.

Quantity Reward. The Regulator then adds a claim-count reward proportional to the number of deduplicated claims n , capped at a target n_{target} :

$$r^{\text{cnt}} = \lambda_{\text{cnt}} \cdot \min\left(\frac{n}{n_{\text{target}}}, 1\right), \quad (6)$$

with $\lambda_{\text{cnt}} = 0.5$ so that factual consistency remains the dominant signal. Because n counts only deduplicated claims, failure modes (1) and (2) are addressed jointly: repetition neither inflates n nor contributes to r^{cnt} .

Quality Penalty. To address failure mode (3), we construct a degraded image $\tilde{I}$ by applying a heavy Gaussian blur, grayscale conversion, and horizontal flipping to I . It then re-answers each question v_i in $\mathcal{Q}^*$ with the degraded image:

$$\tilde{a}_i = \text{Ans}(\tilde{I}, v_i). \quad (7)$$

If $\tilde{a}_i$ remains consistent with the original reference answer a'_i , the claim is marked trivial, i.e., answerable without meaningful visual content. The Regulator penalizes trivial proportions:

$$r^{\text{qual}} = -\lambda_{\text{qual}} \sum_{i=1}^n \mathbf{1}[\mathcal{R}(\tilde{a}_i, a'_i) = 1], \quad (8)$$

pushing the Solver toward responses that require genuine image grounding.

Example. In the airport-scene example, the claim “There are some objects in the image” yields the blurred-image answer “Yes,” which is consistent with the reference answer; accordingly, $\mathbf{1}[\mathcal{F}(\tilde{a}_i, a'_i) = 1] = 1$ and this claim incurs a quality penalty. By contrast, “The yellow dog is on the left of the black dog” cannot be reliably answered from the degraded image, no penalty is applied.

3.5 Checker: Image-Grounded Verification

The Checker receives the filtered claim set $\mathcal{Q}^*$ output by the Regulator and the original image I , with no access to $\mathbf{y}$. For each question $v_i \in \mathcal{Q}^*$, it generates an image-grounded reference answer:

$$a'_i = \text{Ans}(I, v_i), \quad (9)$$

and verifies consistency with the claim-implied answer via majority voting over K independent runs:

$$z_i = \mathbf{1}\left[\sum_{k=1}^K z_i^{(k)} > \frac{K}{2}\right], \quad z_i^{(k)} = \mathcal{F}(a_i, a'_i). \quad (10)$$

The resulting consistency reward is

$$r^{\text{cons}} = \frac{1}{n} \sum_{i=1}^n z_i, \quad (11)$$

measuring the fraction of claims in $\mathcal{Q}^*$ that are supported by image-grounded evidence.

Final Reward. Combining the Checker and Regulator signals, the total reward used for GRPO training is Eq. equation 2. The three terms are complementary: r^{cons} enforces factual grounding; r^{cnt} promotes visual coverage; and r^{qual} filters out image-agnostic claims. Together, the multi-agent pipeline provides a reliable, annotation-free reward signal for scalable hallucination mitigation in VLMs.

4 Experiment

We first outline the experimental setup, the datasets, baselines, and training parameters. We then assess V-MARCH across diverse benchmarks. Finally, we conduct ablation studies to validate the contribution of different components.

Models, Datasets, and Baselines. We evaluate V-MARCH on three sizes of Qwen2.5-VL (3B, 7B, and 32B) (Wang et al., 2024). For the main experiments, we primarily report results on Qwen2.5-VL-7B to ensure fair comparison with baselines at the same model scale. We compare against nine baselines across three categories: (1) the **Base Model** without any post-training; (2) four **self-evolving** models that require no image labels or ground-truth annotations, including **Vision-Zero** (Wang et al., 2026a) (CLEVR, Chart, and Real) and **EvoLMM** (Thawakar et al., 2025); and (3) four models post-trained via RLVR on human-labeled data: **R1-OneVision** (Yang et al., 2025b), **VLAA-Thinker** (Chen et al., 2025), **MM-Eureka** (Meng et al., 2025), and **OpenVLThinker** (Deng et al., 2026). For training, we sample 32K examples from the GQA training split (Hudson and Manning, 2019). For evaluation, we report results on hallucination relevant benchmarks including AMBER (Wang et al., 2023), POPE (Li et al., 2023), VSR (zero-shot) (Liu et al., 2023a), and MME (Fu et al., 2026). To provide a more comprehensive and realistic assessment across diverse capabilities, we further conduct extensive evaluation on A-OKVQA (Schwenk et al., 2022), Real-WorldQA (xAI, 2024), GQA (test) (Hudson and Manning, 2019), BLINK (Fu et al., 2024), Natural-Bench (Li et al., 2024a), SEEDBench (Li et al., 2024b), MMBB (Sun et al., 2025), and MM-Bench (Liu et al., 2024b).

We implement V-MARCH using the Verl framework (Sheng et al., 2025), and evaluate all models using the VLMEvalKit (Duan et al., 2024). We train with a learning rate of 2×10^{-6} , a batch size of 128, and a group size of 8. Further implementation details are provided in Appendix A.1.

4.1 Main Results

Mitigating Hallucination. Table 1 reports results on hallucination-related benchmarks. V-MARCH achieves the best average score of 88.54, outperforming all baselines by a clear margin and improving over the base model by 3.30 points. The gains are particularly pronounced in spatial and re-

Table 1: Performance comparison between V-MARCH and baselines on multiple hallucination datasets. AMBER evaluates attribute recognition, object existence, and spatial relation understanding. MME assesses color, position, and scene perception. POPE measures object hallucination, and VSR evaluates visual-spatial reasoning. **Bold** denotes the best result and underline denotes the second best. Δ indicates the improvement over base model.

Method	AMBER				MME			POPE	VSR	Avg.
	Attribute	Existence	Relation	Accuracy	Color	Position	Scene	—	—	—
Qwen2.5-VL-7B	84.45	94.90	75.66	85.01	90.00	82.50	77.50	86.51	90.64	85.24
VisionZero(CLEVR)	85.28	94.62	78.67	<u>86.19</u>	90.00	82.50	79.00	86.78	<u>91.96</u>	<u>86.11</u>
VisionZero(Char)	84.65	95.04	74.88	<u>84.86</u>	90.00	82.50	77.50	86.23	<u>90.82</u>	<u>85.16</u>
VisionZero(RealWorld)	84.49	94.92	75.90	85.11	90.00	80.00	77.13	86.41	90.21	84.91
EvoLMM	<u>85.29</u>	94.23	77.28	85.60	<u>92.50</u>	81.67	<u>79.75</u>	87.29	91.35	<u>86.11</u>
R1-Onevision	79.39	87.02	<u>81.43</u>	82.61	82.50	60.83	<u>74.50</u>	82.42	72.73	78.16
VLAA-Thinker	84.87	<u>95.29</u>	75.36	85.17	92.50	<u>85.00</u>	77.13	86.19	90.52	85.78
MM-Eureka	85.02	94.62	75.48	85.04	92.50	82.50	77.50	85.71	90.91	85.48
OpenVLThinker	81.27	95.55	75.06	83.96	92.50	75.00	71.13	79.25	86.12	82.20
V-MARCH	86.37	94.58	84.50	88.48	95.00	86.67	82.25	<u>87.04</u>	92.03	88.54
Δ	$\uparrow 1.92$	$\downarrow 0.32$	$\uparrow 8.84$	$\uparrow 3.47$	$\uparrow 5.00$	$\uparrow 4.17$	$\uparrow 4.75$	$\uparrow 0.53$	$\uparrow 1.39$	$\uparrow 3.30$

Table 2: Performance comparison between V-MARCH and baselines on multiple open-ended VQA benchmarks. A-OKVQA (A-OK) evaluates knowledge-grounded visual question answering. RealWorldQA (RWQA) measures understanding of real-world visual scenes QA.

Method	A-OK	GQA	RWQA	Avg.
Qwen2.5-VL-7B	86.38	60.46	67.45	71.43
VZ(CLEVR)	86.81	<u>60.61</u>	67.97	71.80
VZ(Char)	85.76	60.28	67.97	71.34
VZ(RealWorld)	86.29	60.35	67.97	71.54
EvoLMM	<u>86.90</u>	60.59	<u>69.41</u>	<u>72.30</u>
VLAA-Thinker	85.68	60.34	65.49	70.50
MM-Eureka	85.50	60.25	65.10	70.28
V-MARCH	87.60	60.92	70.98	73.16
Δ	$\uparrow 1.22$	$\uparrow 0.45$	$\uparrow 3.53$	$\uparrow 1.74$

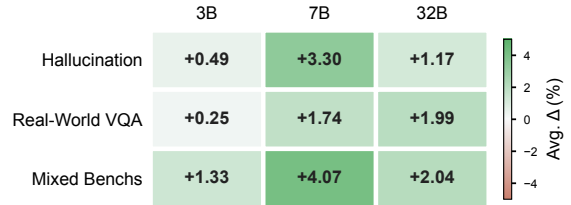


Figure 4: Average improvements ($\Delta\%$) of V-MARCH over the Qwen2.5-VL base model across three model scales and three benchmark categories. All cells are positive, demonstrating consistent gains at every scale.

lational understanding: V-MARCH improves Relation recognition by 8.84 points and Position perception by 4.17 points over the base model. The color dimension and attribute recognition also see substantial boosts of 5.0 and 1.92 points, respectively. These results demonstrate that V-MARCH provides an effective and reliable approach to hallucination mitigation without ground-truth supervision.

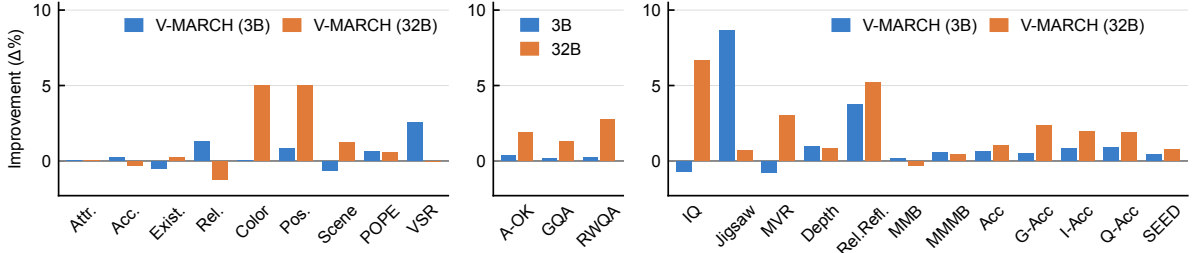
Strong Performance on Real-World VQA. In Table 2, V-MARCH achieves the best performance on all three benchmarks, yielding an average score of 73.16 and surpassing the strongest baseline, EvoLMM, by 0.86 points. The most substantial gain is observed on RealWorldQA (+3.53 over the base model), which requires grounded understanding of complex real-world visual scenes, suggesting that V-MARCH develops more robust visual grounding beyond template-driven or synthetic training signals. Taken together, these results con-

firm that V-MARCH yields consistent and transferable improvements on real-world VQA without sacrificing generalization across task types.

Comprehensive Gains on Multiple Mixed Benchmarks. Table 3 presents results across a diverse set of general-purpose benchmarks spanning low-level visual perception, compositional reasoning, multi-image understanding, and holistic multimodal understanding. V-MARCH achieves the highest average score of 58.29, surpassing the previous best (EvoLMM, 56.77) by 1.52 points and improving over the base model by 4.07 points. In BLINK, V-MARCH achieves the best or second-best score on all five subtasks. On NaturalBench, which evaluates compositional reasoning through stringent group-, image-, and question-level accuracy, V-MARCH again leads overall with improvements across all four metrics. On the broader benchmarks MMBench and MMBench-Multi, V-MARCH achieves scores of 81.61 and 84.34, respectively, along with the highest SEEDBench score of 76.95, demonstrating well-rounded capability across standard multimodal evaluation suites.

Table 3: Performance comparison between V-MARCH and baselines on multiple general multi-task benchmarks.

Method	MMB	MMMB	BLINK					NaturalBench				SEED	Avg.
	–	–	IQ	Jigsaw	MVR	Depth	Rel.Refl.	Acc	G-Acc	I-Acc	Q-Acc	–	–
Qwen2.5-VL-7B	80.62	82.83	0.67	46.67	33.83	71.77	29.85	77.97	31.79	60.39	57.89	76.36	54.22
VZ(CLEVR)	80.99	82.93	1.33	51.33	24.06	70.97	27.61	78.18	32.58	60.76	58.37	76.48	53.80
VZ(Chart)	80.16	82.98	0.67	36.67	32.33	69.35	22.39	78.09	31.63	60.50	58.03	76.48	52.44
VZ(RealWorld)	80.42	83.13	0.00	44.00	<u>39.85</u>	73.39	27.61	78.08	32.26	60.50	58.18	<u>76.59</u>	54.50
EvoLMM	80.88	<u>83.48</u>	<u>4.67</u>	62.00	39.10	<u>75.00</u>	<u>31.34</u>	78.07	31.74	60.32	58.16	76.48	<u>56.77</u>
R1-Onevision	69.61	80.96	0.00	56.00	29.32	<u>28.23</u>	11.19	60.91	8.68	28.13	26.26	69.91	39.10
VLAA-Thinker	80.10	82.27	1.33	35.33	20.30	71.77	28.36	78.33	32.84	61.08	58.66	76.54	52.24
MM-Eureka	80.62	83.23	1.33	32.00	12.03	76.61	15.67	78.37	<u>33.00</u>	61.29	58.71	76.42	50.77
OpenVLThinker	75.12	79.14	0.67	<u>56.67</u>	20.30	51.61	20.15	66.59	16.95	37.95	36.42	72.25	44.48
V-MARCH	81.61	84.34	9.33	<u>56.67</u>	41.35	79.03	38.06	78.53	33.32	<u>61.16</u>	59.18	76.95	58.29
Δ	$\uparrow 0.99$	$\uparrow 1.52$	$\uparrow 8.67$	$\uparrow 10.00$	$\uparrow 7.52$	$\uparrow 7.26$	$\uparrow 8.21$	$\uparrow 0.55$	$\uparrow 1.53$	$\uparrow 0.76$	$\uparrow 1.29$	$\uparrow 0.59$	$\uparrow 4.07$

Figure 5: Per-metric performance improvements ($\Delta\%$) of V-MARCH over the respective Qwen2.5-VL-3B and Qwen2.5-VL-32B base models, across (a) Hallucination, (b) Real-World VQA, and (c) Mixed Benchmarks.

4.2 Model Generalization

Consistent gains across all scales. To evaluate whether V-MARCH generalizes across model scales, we apply the identical training recipe to Qwen2.5-VL-3B and 32B without any scale-specific modifications, and measure per-metric improvements relative to the respective base models. As summarized in Figure 4, every cell of the 3×3 grid is positive, demonstrating that the claim-level self-verification signal benefits models regardless of capacity. The 32B model improves substantially across all three categories ($+1.17$, $+1.99$, $+2.04$), with particularly strong per-metric gains of $+5.00$ on both MME/Color and MME/Position shown in Figure 5. Even the 3B model benefits consistently: it achieves $+8.67$ on BLINK/Jigsaw and $+2.52$ on VSR zero-shot, yielding a $+1.33$ average improvement on Mixed benchmarks. These results confirm that V-MARCH generalizes reliably from 3B to 32B, without any scale-specific adaptations.

4.3 Ablation Studies

VCR Analysis. To assess the contribution of each VCR component, we ablate them individually and report results in Table 4. **Dedup.** proves most critical for the Hallucination benchmark (-1.92): without semantic deduplication, the model exploits

Table 4: Ablation study of the VCR: claim deduplication (Dedup.), claim quantity reward (R_{cnt}), and claim quality penalty (R_{qual}). Each variant removes one component at a time; Decrease denotes the performance drop relative to the full V-MARCH method.

Method	Hallucination	VQA(s)	Mixed(s)
V-MARCH	88.54	73.16	58.29
— w/o Dedup.	86.63	72.03	57.86
Decrease	-1.92	-1.13	-0.43
— w/o R_{cnt}	86.83	71.53	55.45
Decrease	-1.71	-1.63	-2.85
— w/o R_{qual}	87.83	72.30	57.01
Decrease	-0.71	-0.87	-1.28

near-identical claims to inflate the consistency reward without genuinely improving perceptual accuracy, a form of reward hacking that degrades grounded perception. R_{cnt} exerts the broadest influence, yielding the largest drops on Mixed (-2.85) and notable degradation on Real-World VQA (-1.63) and Hallucination (-1.71); this pattern suggests that incentivizing sufficient claim diversity is essential for comprehensive visual coverage across task types. R_{qual} produces comparatively modest but consistent declines across all three categories (-0.71 , -0.87 , -1.28), confirming its complementary role in steering the Solver away from image-agnostic claims toward genuinely visually grounded responses. Taken together, these

Table 5: Comparison of claim-level and response-level self-verification on the three benchmark categories. Decrease denotes the performance drop to the claim-level.

Method	Hallucination	VQA(s)	Mixed(s)
Claim-Level	88.54	73.16	58.29
Response-Level	87.70	72.51	56.99
Decrease	-0.84	-0.66	-1.30

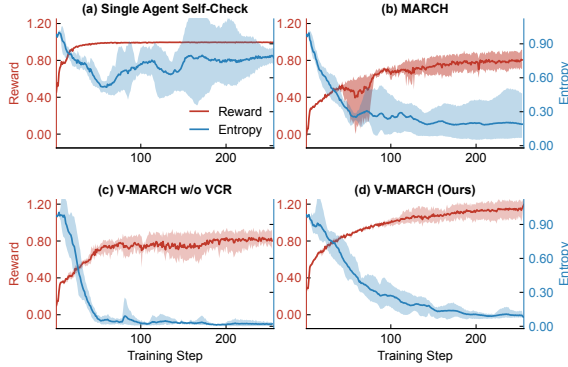


Figure 6: Reward (red, left axis) and entropy (blue, right axis) curves over 256 training steps for (a) Single Agent SelfCheck, (b) V-MARCH w/o VCR, and (c) V-MARCH. The comparison demonstrates that claim-level verification with VCR is necessary for stable, non-degenerate training.

results demonstrate that each VCR component addresses a distinct failure mode, and their combination is necessary for robust, reward-stable training. Overall, all three components contribute positively and consistently, with claim quantity yielding the largest overall gain and claim deduplication being most critical for hallucination mitigation.

Granularity of Self-Verification. A key design choice in V-MARCH is the use of *claim-level* self-verification. To empirically validate the superiority of our decomposed approach, we implement a response-level baseline that replaces the multi-stage claim pipeline. Further implementation details are provided in Appendix A.4. As shown in Table 5, claim-level verification consistently outperforms its response-level counterpart across all three benchmark categories. These results demonstrate that decomposing verification into atomic claims enables more precise reward signals during training, leading to more hallucination-resistant models.

Training Dynamics. Figure 6 compares training stability across four configurations in a 2×2 analysis designed to isolate the contributions of VCR and claim-level reward structure. V-MARCH

Table 6: Wall-clock time per training step, along with agent configuration (trained/total) and relative overhead compared to Single Agent SelfCheck (GT).

Method	Time / Step (s)	#Agents	Overhead
SelfCheck (GT)	171.24	1/1	$\times 1.00$
SelfCheck	180.02	1/1	$\times 1.05$
EvoLMM	360.04	2/2	$\times 2.10$
V-MARCH	281.21	1/4	$\times 1.56$

pipeline is the only configuration that achieves both monotonically increasing reward and controlled entropy decay above 0.1, reflecting genuine policy improvement rather than exploitative shortcutting. This training-time stability, conferred by the claim quantity reward R_{cnt} and quality penalty R_{qual} of VCR, directly underlies the performance advantage V-MARCH demonstrates across all benchmarks.

Computational Overhead. The multi-stage claim pipeline introduces additional inference overhead relative to single-agent baselines. Table 6 reports the wall-clock time per training step for each method. V-MARCH runs four agents to execute the claim extraction, QA generation, image answering, and answer checking stages concurrently, resulting in a per-step time of 281.21 s—a $1.56 \times$ increase over SelfCheck (180.02 s). Given that V-MARCH achieves consistent and substantial performance gains across all evaluation dimensions, this overhead represents a favorable trade-off between compute cost and reward quality.

5 Conclusion

We presented V-MARCH, a visual multi-agent self-verification framework that eliminates the need for ground-truth annotations by decomposing model responses into atomic visual claims and rewarding their consistency with the source image. The central contribution, Visual Claim Regulation, addresses two failure modes that undermine claim-level verification in practice—reward hacking through claim repetition and the generation of image-agnostic claims—and proves necessary for stable, non-collapsing training dynamics. Across hallucination, real-world VQA, and mixed benchmarks, V-MARCH consistently outperforms both response-level and claim-level baselines, and scales reliably from 3B to 32B parameters without modification. We hope this work encourages broader adoption of structured, annotation-free self-verification as a practical path toward more grounded and trustworthy vision-language models.

Limitations

V-MARCH introduces non-trivial inference overhead: the four-agent pipeline incurs approximately $1.56\times$ the wall-clock cost of a single-agent baseline, a burden that compounds at larger model scales and may limit applicability in latency-sensitive settings. Training is conducted on GQA, a standard visual question answering dataset, so performance on visually distinct domains such as document understanding or medical imaging remains unvalidated. The decomposition quality of the Proposer depends directly on the base model’s instruction-following capability, meaning that weak base models may yield malformed or underspecified claim sets that degrade the verification signal. Finally, because the Proposer and Checker are frozen throughout training, the fidelity of the verification loop is ultimately bounded by the visual grounding ability of the off-the-shelf model serving those roles.

Ethical Considerations

This work aims to improve the factual reliability of vision-language models, with potential benefits in high-stakes application domains such as medical image interpretation, legal document analysis, and accessibility tools for visually impaired users. All experiments rely exclusively on publicly available academic datasets and pretrained models; no new data were collected, no human subjects were involved, and no personally identifiable information was processed. As with any advance in the fidelity of generative models, more accurate visual descriptions could in principle be misused to produce more convincing multimodal misinformation, though the primary intent and expected application of this work is the reduction of model hallucination. Finally, the multi-agent verification pipeline incurs a modest additional computational cost (approximately $1.56\times$ that of a single-agent baseline), which carries a proportionally larger carbon footprint; we encourage future work to explore more efficient verification architectures.

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A Appendix

Organization. In this appendix, we provide additional details and descriptions that are not included in the main paper. Specifically, §A.1 presents full implementation details including training hyperparameters and prompt templates, and §A.2 provides additional information on datasets and evaluation protocols.

A.1 Implementation Details

Training Parameters. We train V-MARCH using GRPO (Shao et al., 2024) on top of the VeRL framework (Sheng et al., 2025). The key hyperparameters are summarized in Table 7.

Table 7: Key training hyperparameters used in all experiments.

Hyperparameter	Value
<i>Optimization</i>	
Learning rate	2×10^{-6}
LR schedule	Cosine decay
LR warmup steps	2
LR decay steps	350
Weight decay	0.1
Gradient clipping	1.0
<i>Training</i>	
Training algorithm	GRPO
Global batch size	64
Rollouts per prompt (n)	8
Training epochs	5
Max prompt length	8,192 tokens
Max response length	2,048 tokens
Loss aggregation	Token mean
<i>PPO / GRPO</i>	
Advantage estimator	GRPO
KL coefficient	0.02
KL loss coefficient	0.0
Clip ratio (low / high)	0.2 / 0.2
Entropy coefficient	0.0
<i>Generation</i>	
Temperature	1.0
Top- p	1.0
Top- k	-1 (disabled)
GPU memory utilization	0.6
<i>Parallelism (7B)</i>	
Tensor parallel size	2
Pipeline parallel size	2
Generation tensor parallel	2

Prompt Templates. Our reward pipeline consists of four chained prompts applied sequentially to evaluate the factual accuracy of model-generated captions with respect to the input image. Given a caption, the pipeline proceeds as follows: (1) CLAIMEXTRACT extracts verifiable atomic claims of type color, position, or quan-

tity; (2) CLAIMTOQA converts each claim into a structured question-answer pair; (3) IMAGEQA answers each question directly from the image; and (4) ANSWERCHECK compares the caption-derived answer against the image-grounded answer to produce a binary correctness signal. The four prompt templates are provided below.

CLAIMEXTRACT: Extract verifiable claims from caption

Extract ONLY verifiable claims explicitly stated in the caption, limited to: color, position, and quantity.

Guidelines: - Only extract claims that are explicit and directly stated in the caption. - Do NOT infer unstated attributes or any information not present in the caption. - Each claim MUST be grounded in an exact text span from the caption as "evidence". - Each claim MUST belong to exactly one of three types: color, position, or quantity. - Each claim MUST clearly associate the attribute with a specific object. - If no valid claim exists, output [].

[Color] Include explicit colors (e.g., "red", "light gray"). Example: evidence: "a black cat" → claim: "The cat is black."

[Position] Allowed: left, right, center, top, bottom only. Format: "The [object] is at the [position] of the image."

[Quantity] Explicit numbers only; exclude vague quantifiers (e.g., "several", "many").

Input: [Caption]: {response}

Output: [{"evidence": "...", "claim": "...", "type": "color/position/quantity"}, ...]

CLAIMTOQA: Convert claims into question-answer pairs

Convert claims into Question-Answer pairs using templates.

Guidelines: - Do NOT rewrite or reinterpret the claim. - The question MUST be directly derived from the claim. - The answer MUST be a substring or direct normalization of the claim.

[Color] Q: What color is the [OBJECT]? A: [color phrase]

[Position] Q: Where is the [OBJECT] in the image? A: left / right / top / bottom / center

[Quantity] Q: How many [OBJECT] are in the image? A: [number]

Input: [Claims]: {caption_claim}

Output: [{"claim": "...", "question": "...", "answer": "..."}, ...]

IMAGEQA: Answer questions grounded in the image

Answer the questions based on the image.

Guidelines: - Answer each question based only on the image. - Color questions: answer with

a color word or phrase only. - Position questions: answer with ONLY one of: left, right, center, top, bottom. - Quantity questions: answer with a number word or numeral only. - If the answer cannot be determined reliably from the image, answer "unknown".

Input: [Questions]: {question}
Output: [{"question": "...", "answer": "..."}, ...]

ANSWERCHECK: Compare caption answer vs. image answer

Compare Answer1 (from caption) and Answer2 (from image).

Rules: - Output 1 if they mean the same thing; output 0 otherwise. - Output 0 if Answer2 is "unknown" or either answer is empty.

Matching guidelines: - Ignore format differences: "3" == "three" - Allow simple variants: "blue" == "dark blue" - Allow position variants: "left" == "on the left" - Otherwise, be strict.

Input: [Answer Pairs]: {check_list}
Output: [{"answer1": "...", "answer2": "...", "is_correct": 1/0}, ...]

Image Augmentation Pipeline. To enhance the diversity of training data and improve the model’s robustness to visual perturbations, we apply a stochastic image augmentation pipeline to input images during training. Each of the seven augmentation operations is applied independently with probability $p = 0.125$, with parameters drawn uniformly from the specified ranges at runtime. The full configuration is summarized in Table 9. The Figure 7 illustrates example augmented images generated by the pipeline. At each training step, every enabled operation is applied sequentially to the input image with its corresponding probability, yielding a stochastically augmented view while preserving semantic content. The augmentation parameters are drawn uniformly from the specified ranges at runtime, ensuring sufficient variation across training samples.

A.2 Dataset and Evaluation Details

We utilize the VLMEvalKit (Duan et al., 2024) evaluation toolkit for our assessments, and the following sections provide detailed information on the evaluation protocols and datasets employed in our experiments.

Evaluation Protocol. All models are served via a local vLLM inference server with tensor parallelism and evaluated using the VLMEvalKit frame-

work. For open-ended and multiple-choice questions requiring semantic judgment, we employ Qwen2.5-32B-Instruct as the judge model, accessed through a dedicated API endpoint. Evaluation is conducted in parallel across benchmarks using 64 concurrent API workers (`-api-nproc 64`) to ensure efficient throughput. Results are logged and persisted to a designated output directory for reproducibility.

Datasets Statistics. Our evaluation suite (v2) comprises 20 benchmarks spanning a diverse range of visual understanding tasks, including perception, reasoning, grounding, and multimodal question answering. Table 8 summarizes the benchmark names, primary task categories, and the number of evaluation samples for each dataset. The total evaluation set consists of 94,637 samples across all 20 benchmarks.

Table 8: Statistics of the 20 evaluation benchmarks used in our experiments (v2 suite). Benchmarks are sorted by number of samples.

Benchmark	#Samples
RealWorldQA (xAI, 2024)	765
A-OKVQA (Schwenk et al., 2022)	1,145
VSR (zero-shot) (Liu et al., 2023a)	1,222
MMMB (en) (Sun et al., 2025)	1,980
MME (Fu et al., 2026)	2,374
COCO (Lin et al., 2014)	5,000
POPE (Li et al., 2023)	5,127
BLINK (circular) (Fu et al., 2024)	5,622
MMBench-TEST (V1.1) (Liu et al., 2024b)	7,299
NaturalBench (Li et al., 2024a)	7,600
GQA (Hudson and Manning, 2019)	12,578
AMBER (Wang et al., 2023)	14,216
SEEDBench (Li et al., 2024b)	14,232
Total	79,160

A.3 Albation Details

Table 9: Image augmentation operations and their configurations.

Operation	Range	Probability
Resize	[0.80, 1.20]	0.125
Brightness	[0.80, 1.20]	0.125
Contrast	[0.80, 1.20]	0.125
Sharpness	[0.00, 2.00]	0.125
Gaussian Blur	[0.00, 1.00]	0.125
Gaussian Noise	[0.00, 0.05]	0.125
Salt-and-Pepper Noise	[0.00, 0.01]	0.125



Figure 7: Example augmented images generated by our pipeline. The original image is transformed using each of the seven augmentation operations independently, as well as all operations combined (bottom-right).

Table 10: Ablation study on the augmentation (*aug*) component. “Decrease” denotes the performance change after removing the augmentation module from V-MARCH. Negative values indicate performance degradation.

Method	AMBER				MME			POPE	VSR	Avg.
	Attribute	Existence	Relation	Accuracy	Color	Position	Scene	—	—	—
V-MARCH	86.37	94.58	84.50	88.48	95.00	86.67	82.25	87.04	92.03	88.54
V-MARCH w/o aug	86.39	94.21	84.07	88.23	92.50	86.67	80.63	87.25	91.76	87.97
Decrease	0.03	−0.37	−0.42	−0.25	−2.50	0.00	−1.63	0.21	−0.26	−0.58

Table 11: Ablation study on the augmentation (*aug*) component for open-ended VQA benchmarks. “Decrease” denotes the performance change after removing the augmentation module from V-MARCH. Negative values indicate performance degradation.

Method	A-OK	GQA	RWQA	Avg.
V-MARCH	87.60	60.92	70.98	73.16
V-MARCH w/o aug	86.81	60.99	68.76	72.19
Decrease	−0.79	0.07	−2.22	−0.98

A.4 Response-Level Self-Verification

In addition to the claim-level self-verification described in the main paper, we also explore a response-level self-verification approach where the

model evaluates the overall factual accuracy of its generated response with respect to the input image. Specifically, we design a response-level reward pipeline that directly assesses the correctness of the entire caption without decomposing it into atomic claims.

RESPONSEJUDGE: Holistic response-level verification

Evaluate the following image description for factual accuracy based on the image.

Scoring Guidelines:

- Score 1.0: Fully accurate and consistent with the image. No hallucinations or factual errors.
- Score 0.5: Partially accurate. Some details are correct but there are minor errors or

Table 12: Ablation study on the augmentation (*aug*) component for general benchmarks. BLINK evaluates low-level visual perception including IQ test, jigsaw, multi-view reasoning, relative depth, and reflectance. NaturalBench measures compositional reasoning with overall accuracy, group, image, and question-level metrics. “Decrease” denotes the performance change after removing the augmentation module from V-MARCH. Negative values indicate performance degradation.

Method	MMB	MMMB	BLINK					NaturalBench				SEED	Avg.
	–	–	IQ	Jigsaw	MVR	Depth	Rel.Refl.	Acc	G-Acc	I-Acc	Q-Acc	–	–
V-MARCH	81.61	84.34	9.33	56.67	41.35	79.03	38.06	78.53	33.32	61.16	59.18	76.95	58.29
V-MARCH w/o aug	81.14	83.99	6.00	58.00	36.84	77.42	34.33	78.37	33.42	61.13	58.76	76.56	57.16
Decrease	−0.47	−0.35	−3.33	1.33	−4.51	−1.61	−3.73	−0.16	0.11	−0.03	−0.42	−0.39	−1.13

Table 13: Ablation study of different data processing components on hallucination benchmarks. “w/o dedup” removes the data deduplication strategy, “w/o count” removes the object-count balancing strategy, and “w/o quality” removes the quality filtering strategy.

Method	AMBER				MME			POPE	VSR	Avg.
	Attribute	Existence	Relation	Accuracy	Color	Position	Scene	–	–	–
Qwen2.5-VL-7B	84.45	94.90	75.66	85.01	90.00	82.50	77.50	86.51	90.64	85.24
Ours	86.37	94.58	84.50	88.48	95.00	86.67	82.25	87.04	92.03	88.54
Ours-nodedup	85.51	94.90	81.25	87.22	92.50	82.50	77.13	86.83	91.81	86.63
Ours-nocount	86.05	94.13	83.11	87.77	90.00	84.17	77.50	87.07	91.69	86.83
Ours-noquality	85.66	94.52	82.69	87.62	95.00	85.83	81.88	86.24	91.04	87.83

Table 14: Ablation study of different data processing components on hallucination benchmarks.

Method	A-OK	GQA	RWQA	Avg.
Qwen2.5-VL-7B	86.38	60.46	67.45	71.43
Ours	87.60	60.92	70.98	73.16
Ours-nodedup	86.64	60.57	68.89	72.03
Ours-nocount	85.68	61.21	67.71	71.53
Ours-noquality	86.81	61.32	68.76	72.30

omissions.
- Score 0.0: Significant hallucinations, contradictions, or largely inconsistent with the image.

Evaluation Criteria:

- Object existence: Are the mentioned objects actually present in the image?
- Attribute accuracy: Are colors, sizes, positions, and other attributes correctly described?
- Actions and relationships: Are the described actions and spatial relationships accurate?
- No fabrication: Does the description avoid mentioning things not present in the image?

Output ONLY a JSON object:

```
{“score”: <0.0–1.0>, “reason”: “brief explanation”}
```

Image Description to Evaluate: {response}

Table 15: Ablation study of different data processing components on hallucination benchmarks.

Method	MMB	MMMB	BLINK					NaturalBench				SEED	Avg.
	–	–	IQ	Jigsaw	MVR	Depth	Rel.Refl.	Acc	G-Acc	I-Acc	Q-Acc	–	–
Qwen2.5-VL-7B	80.62	82.83	0.67	46.67	33.83	71.77	29.85	77.97	31.79	60.39	57.89	76.36	54.22
Ours	81.61	84.34	9.33	56.67	41.35	79.03	38.06	78.53	33.32	61.16	59.18	76.95	58.29
Ours-nodedup	81.61	83.33	3.33	58.67	41.35	79.84	35.82	78.62	34.21	61.63	59.21	76.69	57.86
Ours-nocount	81.09	83.03	1.33	60.67	25.56	76.61	29.85	78.14	33.21	61.05	58.45	76.35	55.45
Ours-noquality	81.66	83.79	7.33	56.00	30.08	77.42	38.06	78.57	33.84	61.63	59.08	76.71	57.01